

Task #4 - Roman Converter - Test Automation

Roman Numeral Converter - Test Suite

integerToRoman

- ✓ TC-1: should convert 1 to "I"
- ✓ TC-2: should convert 9 to "IX"
- ✓ TC-8: should throw an error for 4000 (out of range)
- ✓ TC-9: should convert 01 to "I" (leading zero)
- ✓ TC-10: should convert 0010 to "X" (leading zeros)
- ✗ TC-11: should throw error for invalid input like NaN

AssertionError: expected [Function] to throw an error
at n.<anonymous> (test.js:19:51)

- ✓ TC-19: should throw an error for negative numbers
- ✗ TC-20: should handle invalid input type (string)

AssertionError: expected [Function] to throw an error
at n.<anonymous> (test.js:25:59)

- ✓ TC-21: should convert 4 to "IV"
- ✓ TC-22: should convert 49 to "XLIX"
- ✓ TC-23: should convert 3888 to "MMMDCCCLXXXVIII"
- ✓ TC-28: should handle non-integer numbers
- ✓ TC-30: should convert 3999 to "MMMCMXCIX"
- ✓ TC-33: should convert 144 to "CXLIV"
- ✓ TC-34: should convert 900 to "CM"
- ✓ should throw an error for numbers less than 1

romanToInteger

- ✓ TC-4: should convert "Xii" to 12 (case insensitive)
- ✓ TC-5: should convert "XII" to 12
- ✓ TC-6: should throw error for "MMMM" (4000 out of range)

```
expect(() => romanToInteger('MMMM')).to.throw();
```

- ✓ TC-7: should throw error for numeric input 4000 (invalid type)
- ✓ TC-12: should convert "cv" to 105 (lowercase)
- ✓ TC-13: should throw error for "cvb" (invalid character B)
- ✓ TC-14: should throw error for "cv3" (contains number)
- ✓ TC-15: should throw error for "CV9" (contains number)
- ✓ TC-16: should throw error for "A" (invalid character)
- ✓ TC-17: should throw error for "Z" (invalid character)
- ✓ TC-18: should throw error for "XZZX" (invalid characters)
- ✓ TC-24: should throw error for "IIII" (max 3 repetitions rule)
- ✓ TC-25: should throw error for "VX" (invalid subtraction)
- ✓ TC-26: should throw error for "VL" (invalid subtraction)
- ✓ TC-27: should throw error for null input
- ✓ TC-29: should throw error for empty string
- ✓ TC-31: should throw error for "DCCCC" (non-canonical form)
- ✓ TC-32: should throw error for "CCD" (invalid order)
- ✓ TC-35: should convert "MCMXCIV" to 1994
- ✓ should convert "I" to 1
- ✓ should convert "MMMCMXCIX" to 3999

Error Analysis:

TC-11

The integerToRoman function does not verify that the user input is a valid number. When it receives NaN, the condition `num <= 0` is false, and `num >= 4000` is also false. As a result, it does not exit the loop and keeps returning empty strings instead of throwing an error.

A possible solution to this issue would be to validate that the input is a valid number and not NaN.

TC-20

The function receives a string instead of a number. Currently, it does not validate the data type. When it receives "MMMCMXCIX", JavaScript attempts to perform arithmetic operations on a string, which can lead to unexpected results or no result at all.

The solution I propose is to validate that the variable type is number.